

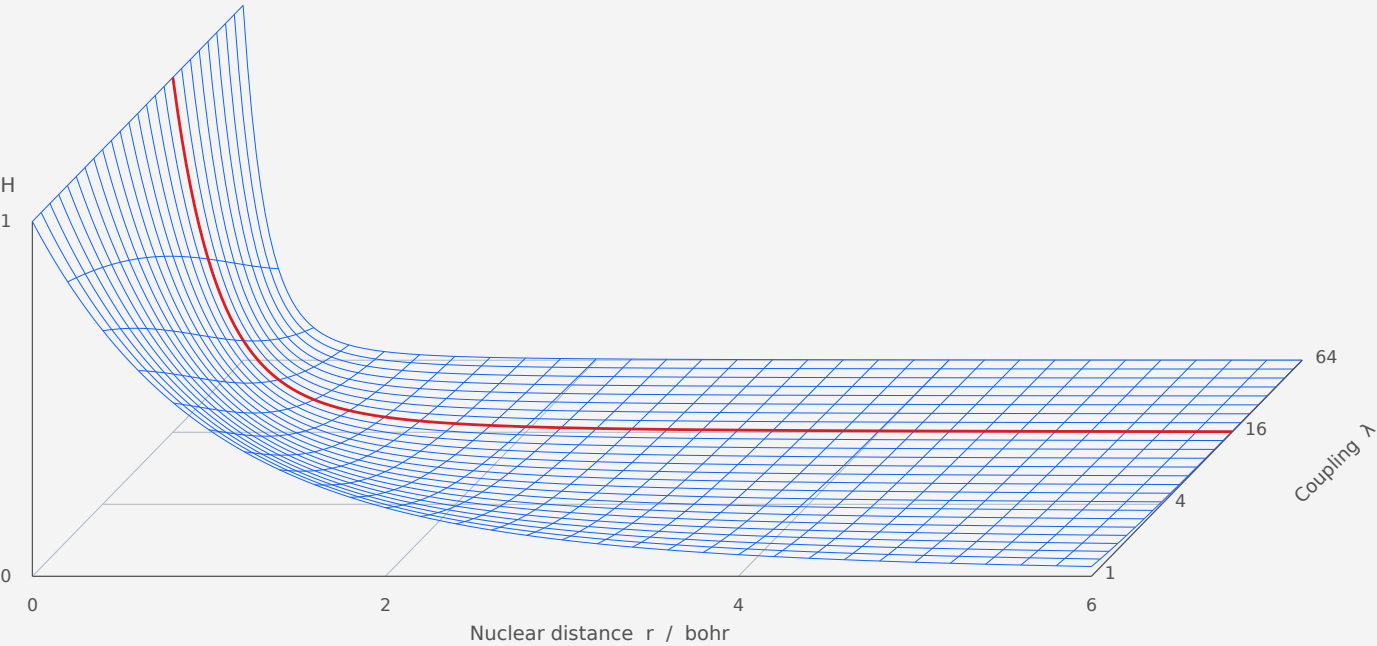
Helium research progress

Toward an exact analytic ground state

PROGRAM / RESULTS / ANALYTICAL PROGRESS / OPEN QUESTIONS

NUCLEAR AXIS / EXACT BOUNDARY PROFILE

One known profile, varying auxiliary coupling.



FROM THE ANALYSIS

The explicit nuclear-axis profile underlies the fractional-power and logarithmic terms in the fixed-energy correction. Blue curves vary the auxiliary coupling; red marks $\lambda = 16$.

$H(t) = 2[1 - (1 + t)e^{-t}]/t^2$, with $t = r\sqrt{2\lambda}$ and the other nuclear distance fixed at one bohr. The plot shows a boundary profile, not a computed helium wavefunction.

[Derivation: attack 9. Spectral analysis and its limitations appear on page 7.](#)

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RESEARCH ATTACKS

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EXACT INVERSE MOMENTS

Open

HELIUM OBJECTIVE

Analytical progress and the remaining gap

We have made substantive analytical progress, but the original helium objective remains open. We can now determine the complete first response to electron repulsion through explicit, convergent integrals of known functions. We have also constructed a convergent equation that selects the physical ground energy and reconstructs its state. The unresolved step is a determining rule that avoids repeated global correlation calculations and demonstrates the reduction required by the research program.

The work has also established why several attractive shortcuts fail. Local cusp conditions and selected moments do not determine the global state. Finite angular truncation cannot close the relevant channel hierarchy. For the actual source in our scalar construction, a finite polynomial kernel sum is impossible, and the fixed-energy correction is not algebraic in the coupling.

THE STRONGEST ANALYTICAL ADVANCES

- We evaluated first-order marginal responses, exact projected connection data, and forced local terms, including a nonzero logarithmic coefficient. An explicit global integral determines the first remaining corner constant.
- We obtained the entire first-order quadrupole function and then the full first-order state, including every angular mode, with no unknown response left inside the defining integrands.
- We proved a route to physical repulsion with an independently selected ground branch, an explicit contraction bound, and a state reconstruction formula.
- We evaluated four inverse spectral moments and derived exact fractional-power and logarithmic terms in the fixed-energy source transform. These determine a sharp spectral moment threshold and the leading cumulative source mass at one spectral edge.

These results go beyond error measurements. Their value is that they determine previously free data, establish global selection, and exclude precisely defined classes of proposed formulas. They do not establish publication novelty; several underlying methods and some early response formulas have clear antecedents.

THE ORIGINAL STANDARD STILL GOVERNS THE OUTCOME

The kickoff program targets nonrelativistic, fixed-nucleus helium at nuclear charge $Z = 2$: the normalized spatial singlet ground state, its energy, every connection datum, and a proof of minimum selection. It also requires a substantive reduction beyond an ordinary systematic solve. The current construction meets several determination and selection obligations, but its total evaluation cost has not met that final research test. [P, 9]

Throughout this report, “first order” means the derivative at zero electron repulsion. A second-order energy coefficient is not the complete helium energy. An estimate, a rigorous error enclosure, and an exact determining formula are identified separately.

[READING ORDER](#) Research narrative 3-8 / Evidence and source map 9

Establishing the first response

Attack 1 separated the exact marginal from the missing correlation

We began with two independent hydrogenic electrons and expanded in the electron-repulsion coupling. The one-electron marginal response was fully evaluated using elementary functions, a logarithm, and an exponential integral. Its normalization was fixed explicitly. Its contribution to the second-order energy coefficient is exactly $-13/432 - (9/32) \log(4/3)$, approximately -0.11100317547 hartree. [1]

The remaining correction has zero marginal for each electron. We proved that it still depends on the difference of the nuclear radii: after the exact marginals are removed, a function of only their sum and the interelectron distance cannot satisfy the response equation. This excludes that particular dimensional reduction, not all possible choices of two variables.

We also generated and independently checked the odd collision terms through degree five. The hierarchy was formal; global convergence and the smooth even sector were not determined. The marginal and odd-sector calculations established a reliable starting point, with acknowledged prior literature.

Attack 2 showed why accurate energies and local constraints were insufficient

A nine-function connected trial tightened the rigorous enclosure of the complete second-order coefficient to $-0.159312 \leq E_2 \leq -0.157458$ hartree. Yet it missed an angular response that had not been imposed during construction by about 40.7 percent. A separate cusp-constrained trial still had a nonzero equation residual. [2]

This was a useful failure. A small energy error and correct averaged cusps did not determine the missing angular structure. We also found a domain obstruction to a naive moment construction: the Coulomb source cannot be repeatedly acted on by the proposed positive-moment operators. Pair-cusp subtraction repairs one level, but a nuclear singularity obstructs the next.

Attack 3 fixed a global coefficient in an exact radial projection

Projecting one electron onto a known hydrogenic 3d state reduced a component of the quadrupole response to one radial equation with an explicit source. Regularity at the nucleus left a coefficient free; decay at infinity fixed it. We evaluated that coefficient as a finite integral and standard hypergeometric functions, then enclosed it by a finite-series certificate. [3]

In the report's unnormalized projection convention, $c_0(1)$ lies between -0.09017580007799561129 and -0.09017580007799561128 , with $c_0(Z) = c_0(1)/Z$. Here "3d" denotes the hydrogenic orbital, not a three-dimensional numerical grid.

This was the first concrete example of the intended global determination mechanism. It fixed one projection, not the full quadrupole. A weighted Coulomb-Sturmian generating equation with a rational source was then derived, but its generating function was not solved.

Selecting and testing global solutions

Attack 4 imposed the physical function space

The generating-equation tests exposed analytic solutions that passed our existing moment checks but were not physical. We resolved the selection problem through the physical function space: an explicit free Green kernel, removal of the known ground mode, and normalization. This produced a globally convergent integral series for the complete first-order response. The sharp contraction factor is $\sqrt[3]{(5/8)}$, and it is $1/3$ in the quadrupole channel. [4]

The theorem controls series truncation in the global L^2 norm. It did not, by itself, evaluate the spatial integrals, control numerical quadrature, or reach full electron repulsion.

Attack 5 separated convergence from achieved numerical accuracy

An unguarded double-precision Bessel-kernel implementation failed 13 of 20 stress cases, including six nonpositive values for a positive kernel. A guarded heat-integral evaluation passed the tested sample. Twenty iterations nearly reproduced a solve on the same grid, but an independent projection check still exposed spatial error. [5]

Keeping the electron-pair cusp exactly and solving for its remainder sharply improved the independent checkpoint. A direct separated solve was several times faster than twenty iterations on the same grid on this host. The theoretical series survived, while the proposed numerical advantage did not.

Attacks 6 and 7 made the response compact and certified a local observable

Separating the overall radius from the ratio of radii made the cusp-subtracted quadrupole response much more compressible. At the same 36-coefficient count, the certified global error improved by about a factor of 197. Enforcing the first nuclear-axis derivative relation then produced a 44-coefficient formula with a stronger global certificate. [6, 7]

Frozen quadrupole formula	Coefficients	Certified weighted L^2 error at $Z = 2$
Attack 5 trial	36	< 0.002775
Attack 6 compact trial	36	< 0.000014107
Attack 7 boundary-constrained trial	44	< 0.000010333

The table bounds the absolute weighted error of the quadrupole component, including its angular polynomial, in $L^2(\mu)$. Here μ is the normalized independent-electron measure. These are not bounds on the entire first-order state.

Attack 7 also certified the relative error of one 5d origin projection below 0.0051 percent. Its signed residual calculation did not read the reference value. An auxiliary one-dimensional calculation supplied the certificate separately from the 44 coefficients of the two-electron formula.

The analytical lesson was specific: ordinary residual-norm control cannot certify that origin projection. We constructed a sequence showing the failure and then derived a dual residual identity that retains the signed information needed for the bound. These are certified approximations to one first-order angular component, not an exact physical helium state.

Determining local and global analytic data

Attack 8 evaluated the local terms and selected a corner constant

The next attack returned from compression experiments to the exact local structure of the first-order quadrupole. We evaluated the forced degree-five term in rational functions, logarithms, arctangent, and Clausen dilogarithms. The previously attempted rational-plus-arctangent form was too narrow. [8]

We then derived the exact coefficient of the forced degree-six logarithm. It is an explicit combination of π , $\log 2$, Catalan's constant, and $\zeta(3)$. Rational bounds prove that it lies between -0.021039051715 and -0.021039051714, so the logarithm cannot be removed. These values use charge $Z = 1$, the negative-response convention Y , and the log-radius gauge fixed in the detailed derivation.

The remaining degree-six angular part was expressed through an explicit angular Green integral. More significantly, the first remaining global amplitude C was determined by an absolutely convergent Whittaker-kernel integral with a fully known source. A separate continuity argument proved that this integral equals the corner coefficient of the physical response.

The numerical estimate $C \approx -0.0087423$ has no certified error interval. Its exact integral determination and its numerical evaluation therefore have different evidentiary status. We also proved that the relevant symmetric angular channels cannot be closed by a finite truncation.

Attack 9 removed growing integration depth from the first response

We first reconstructed the full quadrupole function at all radii. Evaluating one of the source integrations in explicit hypergeometric moments reduced it to a fixed two-dimensional quadrature over frequency and radius. Global selection, absolute convergence, and reconstruction bounds were proved. This determines the complete function, not just a selected origin coefficient. [9]

We then used the closed Coulomb Green function for a single electron, removed its ground-state pole analytically, and evaluated its zero-frequency limit explicitly. Combining these known kernels gives the complete normalized first derivative as a fixed seven-dimensional, pointwise absolutely convergent integral plus the evaluated one-electron terms. Every angular mode is included.

The kernel checks cover its differential equation, free limit, removed pole, angular projections, normalization, negative energy shifts, and charge scaling. We have not numerically evaluated the complete seven-dimensional source integral. The representation is nevertheless an explicit determining formula: its integrand contains no unknown interacting response or unevaluated interacting inverse.

Attack 4 had already determined the complete first derivative through a convergent integral series. Attack 9 replaces that growing integration depth with fixed-dimensional known-function quadratures, extending well beyond attack 8's corner calculation. The finite-coupling reduction remains unresolved. The underlying Coulomb kernels and contour-convolution machinery are established methods; no priority claim is attached to their use here.

Reaching physical electron repulsion

A uniform gap selects the ground branch

Write the Hamiltonian as $H_\lambda = H_0 + \lambda W$, where W is the reciprocal interelectron distance. Zero coupling gives independent electrons; $\lambda = 1$ is the physical repulsion. At $Z = 2$, elementary operator bounds and a simple variational state give

$$-4 \leq E_\lambda \leq -\left(2 - \frac{5\lambda}{16}\right)^2 \leq -\frac{729}{256} < -\frac{5}{2}, \quad 0 \leq \lambda \leq 1.$$

The next min-max level is at least $-5/2$, so the ground branch has a uniform gap of at least $89/256$ hartree. This proves that the selected branch reaches the physical Hamiltonian. It uses no fitted helium energy. An independent construction through overlapping analytic neighborhoods also reaches the endpoint. [9]

The energy is the unique root of an explicit scalar construction

Let p be the normalized independent-electron ground state, and let Q remove its component. Set $g = QWp$. At a trial energy E in the proved bracket, the complement response χ is defined by the inverse of $Q(H_\lambda - E)Q$ acting on g . We then replaced this inverse by a convergent recurrence built from explicit independent-electron kernels.

The scalar condition and normalized reconstruction are

$$\begin{aligned} \mathcal{F}_\lambda(E) &= -Z^2 + \frac{5Z\lambda}{8} - E - \lambda^2 \langle g, \chi(E) \rangle = 0, \\ \psi_\lambda &= \frac{p - \lambda \chi(E_\lambda)}{\sqrt{1 + \lambda^2 \|\chi(E_\lambda)\|^2}}. \end{aligned}$$

Its derivative with respect to E is at most -1 , and the bracket signs and spectral gap select exactly one ground-state root. A rescaled inverse iteration converges uniformly with contraction $q \approx 0.8226269582$. The associated truncation error is explicit. In contrast, the unrescaled Born operator series fails at one bracket endpoint: a hydrogenic test state gives a norm lower bound $86528/64881$, which exceeds one.

Why the original endpoint is still open

Each iteration requires a new globally correlated amplitude. Direct nesting increases the integral dimension; storing and reusing the amplitude requires a controlled representation of the three internal variables. We have not replaced those repeated operations with an evaluated generating rule or demonstrated a computational advantage over established systematic solvers.

Thus the branch, recurrence, and state reconstruction are determined, but the research program's stronger reduction requirement is unmet. The scalar reduction has established Feshbach-Schur antecedents. No new numerical value of the complete interacting energy is claimed in this work.

Exact constraints on any future summation

The actual source excludes finite polynomial and algebraic closure

For a fixed trial energy E below $-5Z^2/8$, define the positive complement kernel $B = A^{-1/2}QWQA^{-1/2}$, where $A = Q(H_0 - E)Q$. The source vector is $\eta = A^{-1/2}g$. Its scalar correction has the positive spectral representation

$$S_E(\lambda) = \langle \eta, (I + \lambda B)^{-1} \eta \rangle = \int \frac{d\mu_E(b)}{1 + \lambda b}.$$

We proved that the actual source measure has infinite support. Consequently, no finite polynomial of kernel applications can yield its exact response, and its transform cannot be rational. This is stronger than merely showing that the operator has infinitely many spectral values. [9]

The nuclear singularities force a fractional power and a logarithm

Four inverse moments $m_{-j} = \int b^{-j} d\mu_E(b)$ were evaluated exactly. They are polynomials in $\tau = (-Z^2 - E)/Z^2$ with rational coefficients and $\log 2$. For example, $m_{-1} = 47Z/280$ and $m_{-2} = Z^2(311\tau + 256)/1225$. The fourth is finite as an energy form, even though the next stronger operator-domain condition fails.

An explicit profile near each nuclear axis, together with a global residual estimate, yields

$$S_E(\lambda) = \sum_{j=1}^4 (-1)^{j-1} m_{-j} \lambda^{-j} + \frac{72\sqrt{\pi}}{35} Z^{9/2} \lambda^{-9/2} - \frac{1024}{245} Z^5 \lambda^{-5} \log \lambda + O(\lambda^{-5}).$$

The nonzero logarithm excludes an algebraic fixed-energy correction. Independent audits checked the profile, projection, singular domains, both coefficients, and the stronger remainder needed to justify the logarithmic term.

Inverse moments are finite exactly for nonnegative orders below $9/2$. A Tauberian argument also determines the cumulative source mass:

$$\mu_E((0, b]) \sim \frac{16Z^{9/2}}{35\sqrt{\pi}} b^{9/2} (b \downarrow 0).$$

No spectral density is assumed. These are exact constraints on the source measure, but they do not determine its interior. The asymptotic expansion holds as the auxiliary coupling tends to positive infinity with E fixed. It is not an expansion of the physical ground energy E_λ and does not evaluate the correction at $\lambda = 1$.

The current test and the remaining decision

Attack 10 is testing finite differential closure

The latest route asks whether the fixed-energy correction could satisfy one finite linear differential equation with polynomial coefficients in the coupling, a class often called D-finite. Infinitely many distinct poles would exclude that class.

For the attractive family $H_0 - tW$, we established an infinite discrete spectral ladder in the complement operator above the edge $b^* = 1/(2\sqrt{-E})$. These eigenvalues accumulate at b^* . The essential spectrum is exactly $[0, b^*]$, including within the exchange-symmetric S sector. The proof uses an escaping bound electron pair whose center still feels an attractive Coulomb tail, together with the cluster threshold and the min-max principle. [10]

The missing step is source visibility. An operator pole appears in the scalar correction only when its eigenstate couples to the specific source. We derived the exact condition for a state to be invisible and proved a useful local result: when attraction is first switched on, the source couples to every hydrogenic singlet S excitation of the form $1s\ n s$, $n \geq 2$.

That local result does not prove nonzero weights on infinitely many members of the later complement ladder. Threshold changes and continuation through crossings require an additional argument. We therefore do not claim a D-finite impossibility theorem, even generically in E. This remains a promising but incomplete test, not a successful summation.

Where the program now stands

Research target	Current position
Determine the complete first-order state	Explicit, absolutely convergent known-function integrals obtained
Fix local and global connection data	Full first-order response fixed; higher individual connection coefficients not separately evaluated
Reach the physical ground branch	Uniform gap, unique scalar root, and convergent reconstruction proved
Produce a reduced exact finite-coupling determination	Repeated global amplitude operations remain
Demonstrate a total computational advantage	Not established

The next analytical step should remove an unknown

The clearest current target is the global source transform or an equivalent summation rule. Four exact moments and one spectral-edge law constrain it, but do not specify it. A viable candidate must determine its remaining data, reach $\lambda = 1$ with controlled evaluation, and account for the cost of coefficient generation and verification.

The spectral-ladder test should be pursued only through a proof of source visibility or a precise obstruction to that proof route. More energy digits or another unconstrained series would not resolve the remaining issue. There is no proof that every exact analytic helium representation is impossible, and the present failures must not be read that broadly.

Evidence and source map

The research records below contain the derivations, frozen coefficients, exact certificates, numerical diagnostics, and independent audits underlying this summary. All paths are relative to the Helium folder. The original program and the research records are dated 12 September 2026; the attack numbers identify their sequence.

Program and research records

[P] Helium_Research_Program_and_Literature_Guide.pdf, especially pages 3-5 and 16. Physical target, exactness standard, cost requirement, and relation to established methods.

[1] research/first_attack/derivation.md and README.md. Exact marginal, connected-response obstruction, formal odd sector, and initial coefficient bounds.

[2] research/second_attack/report.md. Connected trial, unused angular test, sharper coefficient enclosure, and source-domain obstruction.

[3] research/third_attack/report.md and generating_equation.md. Exact 3d projection, certified connection coefficient, and unsolved generating equation.

[4] research/fourth_attack/report.md. Physical selection, explicit kernel series, sharp contraction bounds, and unphysical analytic modes.

[5] research/fifth_attack/report.md. Numerical failure tests, exact cusp subtraction, same-grid cost comparison, and separate continuum certificate.

[6] research/sixth_attack/report.md. Frozen compact quadrupole formulas, exact residual certificates, and independent 5d validation.

[7] research/seventh_attack/report.md. Boundary-constrained formula, origin-projection certificate, and the failure of norm-only origin control.

[8] research/eighth_attack/report.md. Evaluated degree-five term, exact degree-six logarithm, global corner integral, and finite angular-channel obstruction.

[9] research/ninth_attack/README.md, physical_scalar.md, and source_nonclosure.md. Complete first-order kernels and physical-coupling construction. The subfolders agent_inverse_moments, agent_edge_asymptotic, and agent_edge_domains contain the moment formulas, spectral-edge proof, and independent audits.

[10] research/tenth_attack/README.md and hydrogenic_visibility.md, with linked spectral-ladder and source-visibility notes. Latest operator theorem, exact local coupling calculation, and unresolved continuation issue.

Selected mathematical antecedents

The original program contains the broader literature guide. The early marginal and local odd-sector calculations have antecedents in Hall, Jones and Rees; White and Brown; and later work discussed in the first-attack source audit. A term-by-term priority comparison of the newer coefficient formulas has not been completed.

- Hostler and Pratt (1963), Coulomb Green's Function in Closed Form. DOI [10.1103/PhysRevLett.10.469](https://doi.org/10.1103/PhysRevLett.10.469). Basis for the explicit Coulomb kernels.
- Simon (1982), Schrödinger semigroups. DOI [10.1090/S0273-0979-1982-15041-8](https://doi.org/10.1090/S0273-0979-1982-15041-8). Semigroup estimates and continuity used in global connection analysis.
- Dusson, Sigal and Stamm (2021), The Feshbach-Schur map and perturbation theory. [arXiv:2105.02058](https://arxiv.org/abs/2105.02058). Context for scalar reduction, root selection, and inverse iteration.
- Langer and Woracek (2025), Karamata's theorem for regularized Cauchy transforms. DOI [10.1017/prm.2023.128](https://doi.org/10.1017/prm.2023.128), Theorem A.7 and Remark A.8. The Tauberian theorem used for cumulative spectral mass.

The calculations use no interacting reference energy, density, or wavefunction as a determining input. Numerical comparisons and proof certificates retain their separate roles. Earlier local labels such as “complete” do not signify completion of the overall helium program.